

Goldbach-type representations with prescribed Liouville signs

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Abstract

Let λ be the Liouville function. For every sufficiently large integer N , each of the four ordered sign patterns $(\lambda(a), \lambda(b))$, with $a, b \geq 1$ and $a+b = N$, occurs $N/4 + O(N(\log N)^{-c})$ times, where one may take $c = 10^{-200}$. Equivalently, $\sum_{1 \leq n < N} \lambda(n)\lambda(N-n) = O(N(\log N)^{-c})$; this proves, for the Liouville function, the conjecture of Corradi and Katai that the sum is $o(N)$. The proof combines the finite graph and residue-comparison estimates of OpenAI's Lean 4 development of two-point correlations, the Dirichlet zero-free half-plane $\operatorname{Re} s > 7/8$ from family 003 of the same library, and the short-progression variance bounds of Klurman–Mangerel–Teräväinen; the deduction from these inputs is carried out in the paper and is formalized in Lean 4, including the exponent 10^{-200} , with one published estimate as hypothesis. A quotient-coordinate construction fixes the graph multiplier while the total N remains in the progression modulus. Inserting the same half-plane into published results of Pintz, Chen–Gupta–Li, Montgomery–Vaughan and Martin gives further consequences: the exponent $3/5$ for the exceptional set in the binary Goldbach problem, which Pintz proved under such a zero-free hypothesis; the bound $O_\varepsilon(q^{7/3+\varepsilon})$ for the least prime in a reduced class modulo q ; the exponent 8 in the polylogarithmic bound for the least quadratic non-residue; and the bound $O((\log p)^{24})$ for the least prime primitive root modulo p .

1 Statement and context

All logarithms are natural. Write $\lambda(n) = (-1)^{\Omega(n)}$ for $n \geq 1$, where prime factors are counted with multiplicity, and put

$$C(N) = \sum_{1 \leq n < N} \lambda(n)\lambda(N-n),$$
$$A_{\epsilon_1, \epsilon_2}(N) = \#\{(a, b) \in \mathbb{N}_{>0}^2 : a+b = N, \lambda(a) = \epsilon_1, \lambda(b) = \epsilon_2\}.$$

The pairs in this definition are ordered.

Theorem 1.1. *There are absolute constants $C_0, N_0 > 0$ such that, for every integer $N \geq N_0$ and every $(\epsilon_1, \epsilon_2) \in \{-1, 1\}^2$,*

$$\left| A_{\epsilon_1, \epsilon_2}(N) - \frac{N}{4} \right| \leq C_0 \frac{N}{(\log N)^{10^{-200}}}. \quad (1)$$

Moreover,

$$|C(N)| \leq C_0 \frac{N}{(\log N)^{10^{-200}}}. \quad (2)$$

The inputs are the named theorems of families 003 and 007 in [23], at commit

Family 003 supplies the Dirichlet zero-free half-plane; family 007 supplies the finite closed-word estimates, residue comparisons and spectral transfer. At this commit these Lean 4 theorems compile with only the axioms `propext`, `Classical.choice` and `Quot.sound`. Inputs 2.1, 3.2–3.4 and 4.2 transcribe the named declarations into mathematical notation. The numerical exponent of Lemma 3.5, the reductions to these inputs and the proof of Theorem 1.1 are written mathematics.

Theorem 1.1, with the exponent 10^{-200} , is also formalized in Lean 4; the files are in the directory `lean` of the repository of this paper. The theorem `paper_theorem_1_1_one_hypothesis` proved there has one hypothesis: the variance estimate of [12] for real multiplicative functions, with the explicit set of moduli of its Proposition 9.4 and on the range of parameters used in Lemma 2.3. It compiles against the library at the commit above with the same three axioms. In that proof the prime-sum estimate in Lemma 2.2 is derived from Input 2.1 by a holomorphic logarithm and the Borel–Carathéodory inequality, the Halász step is the library’s theorem for completely multiplicative functions, and [18, Lemma C.1] for the trivial character is proved from the library’s zeta-function and prime-number estimates. The comparison theorems of the library assert only $\exists A_f \geq 1000$; the bound $A_f \leq 10^6$ of Lemma 3.5 is read off from their source proofs, and the formalization proves these comparisons again with fixed exponents. The ancillary file `ExplicitBudget.lean` checks the polynomial exponents 8457 and 8458 of the degree function. The other Lean files cited below are also ancillary files of this submission.

The final two-point theorems of [22] fix affine forms with positive leading coefficients and non-negative constant terms as the summation length grows, with constants depending on the forms. Here the form $N - n$ varies with the summation length. The argument runs the internal graph estimates at shift/length 1, with [12] and Input 2.1 supplying the variance of λ in progressions modulo N needed for centring.

Theorem 1.1 proves, for the Liouville function, the conjecture of Corradi and Katai that $C(N) = o(N)$, as stated in [14, §1.1] and [13, Conjecture 1]. Those papers give [8] as its source. De Koninck, Germán and Kátai [9] established a conditional result under the existence of infinitely many Siegel zeros; we use the description in [14, §1.1]. Mangerel [14, Theorem 1.2] proved $|C(N)| < N - 1$ for every $N \geq 11$; for $N \geq 2$, equality holds exactly when $N \in \{2, 3, 5, 10\}$.

For sign patterns, Shusterman asked whether every even $N > 2$ is a sum $a + b$ with $\lambda(a) = \lambda(b) = -1$; the question was posted by Pablo on MathOverflow in August 2018 [26]. Under GRH, Mangerel [15, Theorems 1.2–1.3 and Proposition 1.4] proved this for all large even N , and all four patterns for every N with a prime factor exceeding a fixed p_0 . For sufficiently large prime N , each pattern occurs $\gg N \exp(-C(\log \log N)^6)$ times. His Remarks 1 and 9 replace GRH, for the two existence theorems, by a zero-free rectangle $\operatorname{Re} s > 1 - (\log N)^{-\alpha}$, $|\Im s| \leq (\log N)^3$, for nonprincipal characters modulo N , with $0 < \alpha < 3/50$; Input 2.1 supplies this rectangle for large N . CaptainSude [5, Theorem 1.1] gave an elementary proof, with a Lean formalization, for every even $N > 2$. Meng [19, Theorem A] proved that all four patterns occur for every $N \geq 2$ outside $\{2, 3, 4, 5, 6, 9, 10\}$. Pouly [29, Theorem 1.1] proved that, for every fixed $\varepsilon > 0$, each of the four patterns occurs at least $c_\varepsilon N^{1-\varepsilon}$ times for every sufficiently large N . For $G_k(N) = \sum_{a_1+\dots+a_k=N} \prod_i \lambda(a_i)$, with $a_i \geq 1$, Krishnamoorthy [13, Theorems 1–2] proved $|G_k(N)| \ll_{B,k} N^{k-1} (\log N)^{-B}$ for each fixed $k \geq 3$ and $B > 0$, and, for $k = 2$, $\#\{N \leq X : |C(N)| > tN\} \ll_B t^{-2} X (\log X)^{-B}$ for $t > 0$. Theorem 1.1 gives the asymptotic count of all four patterns for every sufficiently large total. Constants and thresholds in this note are chosen existentially; their numerical values are not tracked.

Lemma 1.2. *For every $N \geq 2$ and every pair of signs,*

$$A_{\epsilon_1, \epsilon_2}(N) = \frac{N - 1 + (\epsilon_1 + \epsilon_2) \sum_{1 \leq n < N} \lambda(n) + \epsilon_1 \epsilon_2 C(N)}{4}. \quad (3)$$

Consequently a bound for $C(N)$ with an absolute logarithmic saving, together with such a bound for $\sum_{n < N} \lambda(n)$, gives the four counts with the same smaller saving. Conversely, the four counts imply the corresponding bound for $C(N)$.

Proof. Expand $\frac{1}{4} \sum_{1 \leq n < N} (1 + \epsilon_1 \lambda(n))(1 + \epsilon_2 \lambda(N - n))$. The two linear sums agree under $n \mapsto N - n$. For the converse use $C(N) = \sum_{\epsilon_1, \epsilon_2} \epsilon_1 \epsilon_2 A_{\epsilon_1, \epsilon_2}(N)$; the four main terms cancel. $\square$

Part I

The reflected correlation

2 Zero-free input and progression estimates

Input 2.1 (Family 003, [25, Theorem 1.1]). *For every positive integer q , every Dirichlet character $\chi \pmod{q}$ and every $s \in \mathbb{C}$ with $\operatorname{Re} s > 7/8$, one has $L(s, \chi) \neq 0$, apart from the principal character's pole at $s = 1$.*

The exact Lean name is

`OAI.DirichletCharacter.LFunction_ne_zero_of_seven_eighths_lt_re.`

Its character is Mathlib's `DirichletCharacter` $\mathbb{C} q$, its modulus has the instance `[NeZeroq]`, and its explicit pole hypothesis is $\neg(\chi = 1 \wedge s = 1)$. The companion theorem `OAI.riemannZeta_ne_zero_of_seven_eighths_lt_re` supplies the zeta statement. Mathlib's zeta function is totalized at 1; away from that point the statement is the usual zero-free assertion for zeta.

Put $e(t) = \exp(2\pi it)$ and, for 1-bounded multiplicative functions,

$$\mathbb{D}(f, g; X)^2 = \sum_{p \leq X} \frac{1 - \operatorname{Re}(f(p)\overline{g(p)})}{p}.$$

Lemma 2.2. *There are absolute $C, X_0 > 0$ such that, for all $X \geq X_0$, all $1 \leq q \leq X$, with q integral, and all real Dirichlet characters $\chi \pmod{q}$,*

$$\left| \sum_{n \leq X} \lambda(n) \chi(n) \right| \leq CX (\log X)^{-1/40}. \quad (4)$$

Principal and imprimitive characters are included.

Proof. For a nonprincipal primitive character, truncated Perron at a half-integer $y \geq 3$, with right line $1 + 1/\log y$ and height $T \asymp y^2$, can be moved to $15/16$. Montgomery–Vaughan's local expansion for L'/L [20, Lemma 11.1 and Theorem 10.17] and Input 2.1 give, uniformly on $15/16 \leq \sigma \leq 2$,

$$|L'/L(\sigma + it, \chi)| \ll \log(2q(|t| + 2)).$$

Indeed every zero in the local expansion is at horizontal distance at least $1/16$, and the number of local zeros is bounded by the stated unit-height zero-counting theorem. The vertical integral is $O(y^{15/16} \log(2qT) \log(2T))$; the horizontal integrals and Perron error are smaller. Moving the cutoff by at most one von Mangoldt term therefore gives

$$\psi(y, \chi) \ll y^{15/16} \log^2(2qy), \quad y \geq 2. \quad (5)$$

An induced character differs from its primitive character by Euler-factor terms of total size at most $O(\log(2q)\log(2y))$. Removing prime powers costs $O(\sqrt{y}\log^2(2y))$. Thus (5) also bounds $\sum_{p \leq y} \chi(p) \log p$, for every nonprincipal character, with an absolute constant and without a lower bound on y/q . The range $2 \leq y < 3$ follows by the absolute bound for its finitely many von Mangoldt terms.

Set $Y = (\log X)^{64}$. Partial summation, using $q \leq X$, gives

$$\left| \sum_{Y < p \leq X} \frac{\chi(p)}{p} \right| \ll (\log X)^2 Y^{-1/16} \ll 1.$$

For $f = \lambda\chi$, this and Mertens' theorem yield

$$\mathbb{D}(f, 1; X)^2 = \sum_{p \leq X} \frac{1}{p} + \sum_{p \leq X} \frac{\chi(p)}{p} \geq \log \log X - O(\log \log \log X).$$

For a principal character the second sum is nonnegative, so the same lower bound holds.

Apply Matomäki–Radziwiłł–Tao [18, Lemma C.1, corrected version 3] with its character slot equal to the fixed modulus-one character and its real function slot equal to the varying f . For $1 \leq |t| \leq X$ it gives $\mathbb{D}(f, n^{it}; X) \geq \frac{1}{4} \sqrt{\log \log X} - O(1)$; for $|t| \leq 1$ it gives $\mathbb{D}(f, n^{it}; X) \geq \frac{1}{3} \mathbb{D}(f, 1; X) - O(1)$. Hence, after increasing an absolute X_0 ,

$$M := \inf_{|t| \leq \sqrt{\log X}} \mathbb{D}(f, n^{it}; X)^2 \geq \frac{1}{32} \log \log X.$$

The Halász inequality [12, Lemma 7.4, modulus 1], with $T = \sqrt{\log X}$, bounds $X^{-1} |\sum_{n \leq X} f(n)|$ by

$$C\{(M+1)\exp(-M) + T^{-1/2} + (\log X)^{-1/4}\} \ll (\log X)^{-1/40}.$$

The constants in the invocation of Lemma C.1 are absolute because its character is fixed at modulus one. $\square$

Lemma 2.3. *Fix $0 < \nu \leq 1$. There are $C_\nu, X_\nu > 0$ such that, for every positive integer q , whenever $X \geq X_\nu$, $X \geq H \geq 10q$, and $R = H/q \geq \exp((\log X)^\nu)$,*

$$\int_X^{2X} \sum_{a \bmod q} \left| \sum_{\substack{x < n \leq x+H \\ n \equiv a \pmod{q}}} \lambda(n) \right|^2 dx \leq C_\nu \frac{XH^2}{q} (\log R)^{-1/1000}. \quad (6)$$

The sum is over every residue class, including nonunits.

Proof. Write $\varepsilon = (\log R)^{-1/1000}$. Use an external modulus upper bound which may exceed the actual q . Set

$$T = \log X, \quad \rho = \varepsilon^{11/10}, \quad \kappa = \rho/50, \\ Q_* := \left\lceil \max\{qR^\kappa, H \exp(-T^{2/5})\} \right\rceil, \quad R_* = H/Q_*, \quad S_* = \log R_*.$$

For large X , uniformly in the stated range,

$$q \leq Q_* \leq H/10, \\ \frac{1}{2} \min(R^{1-\kappa}, \exp(T^{2/5})) \leq R_* \leq \min(R^{1-\kappa}, \exp(T^{2/5})). \quad (7)$$

Both arguments of the maximum are $o(H)$ and the first is at least q . The ceiling costs a factor at most two. In particular, with $\nu_1 = \min(\nu, 2/5)$, $S_* \geq T^{\nu_1}/3$ for large X and $S_* \leq T^{2/5}$. The internal parameter lower bound is still $\rho \geq S_*^{-1/200}$: before capping, $S_* \asymp \log R$ and $0.0011 < 1/200$; after capping, $\rho \geq T^{-0.0011}$ while $S_*^{-1/200} \asymp T^{-1/500}$. Also $S_* \gg \rho^{-100}$ in both cases.

The explicit good-modulus set used in [12, Lemma 8.1] is

$$\mathcal{Q}_{X,u,M} = \left\{ q \leq X : \begin{array}{l} L(s, \chi) \neq 0 \text{ for all } \chi \pmod{q} \text{ with } \text{cond } \chi > X^{u_{20}}, \\ \text{Re } s \geq 1 - M \log \log X / \log X, \quad |\Im s| \leq 3X \end{array} \right\}.$$

The deduction of Theorem 1.5 from Proposition 9.4 in version 5 uses $\varepsilon^{1.1}$ in that proposition. Its good-modulus condition is therefore $\mathcal{Q}_{X,\varepsilon^{6.6},\varepsilon^{-88}}$. This is contained in $\mathcal{Q}_{X,\varepsilon^6,\varepsilon^{-80}}$, which is contained in $\mathcal{Q}_{X,\varepsilon^6,\varepsilon^{-36}}$ used for the real-function reduction. The first set has the smaller conductor cutoff and larger zero box. Since

$$\varepsilon^{-88} \frac{\log \log X}{\log X} \leq (\log X)^{-0.912} \log \log X < \frac{1}{8},$$

Input 2.1 makes all three sets equal to all moduli up to X . The reduction from Proposition 9.4 uses Lemma 7.7, which needs $Z = \varepsilon^{-11} \leq (\log X)^{1/20}$ and $q \leq H/10$, both satisfied; Lemma 6.2, the hybrid large sieve valid for all $q, T \geq 1$; and equation (50), which [12] derive from Lemma 14 of Matomäki and Radziwiłł[17], which is reference [32] there. We use this last step as printed. None restricts q beyond $q \leq Q_* \leq H/10$. The real-valued simplification is Corollary 1.6 and its proof. Use Q_* as the external Q in these invocations. The accuracy ε and the actual modulus q remain unchanged.

Here are the parameter margins for the nested invocations in that proof:

Parameter	Margin
$\varepsilon^{1.1}$	$(\log R)^{-0.0011} \geq (\log R)^{-1/200}$
$\varepsilon^{6.6}$	$\geq (\log X)^{-0.0066} > (\log X)^{-1/50}$
$20\varepsilon^{-220} \log \log X / \log X$	$\leq 20(\log X)^{-0.78} \log \log X = o(1)$
ε^{-11}	$\leq (\log X)^{0.011} < (\log X)^{1/20}$
ε^7	$\geq (\log X)^{-0.007} > (\log X)^{-1/13}$

For the internal $\varepsilon^{1.1}$ invocation the last margin has exponent 0.0077, still below $1/13$. We also check the auxiliary prime intervals in Section 9.3 of [12], including its equation (52). Let $m \geq 2$ be the smallest integer with $m^{4m+2} S_*^m > T^{1/2}$. Define

$$\begin{aligned} \log P_1 &= \rho S_*, & \log Q_1 &= S_*; \\ \log P_j &= \rho j^{4j} S_*^j, & \log Q_j &= j^{4j+2} S_*^j \quad (2 \leq j < m), \end{aligned}$$

and $\log P_m = \rho^2 T$, $\log Q_m = \rho T$. For large X , $m \leq b_\nu := 2 + \lceil 1/\nu_1 \rceil$, since $S_* \geq T^{\nu_1}/3$. With $\eta_a = 1/100$, the required inequalities are

$$\begin{aligned} \frac{\log \log Q_j}{\log P_{j-1} - 1} &\leq \frac{\eta_a}{4j^2}, \\ \frac{\eta_a}{j^2} \log P_j &\geq 8 \log Q_{j-1} + 16 \log j \quad (2 \leq j \leq m). \end{aligned} \tag{8}$$

For $j < m$, the first ratio is $O_\nu(\log S_*/(\rho S_*)) = o(1)$, using $S_* \gg \rho^{-100}$; the second comparison gains a factor comparable to $\rho S_* \rightarrow \infty$. For $m = 2$, its defining inequality gives $S_* > T^{1/4}/32$,

so $\log P_1 \geq T^{1/4-11/10000}/32$. Meanwhile $\log P_2 \geq T^{1-22/10000}$ and $\log Q_1 \leq T^{2/5}$; both terminal inequalities follow. For $m \geq 3$, minimality gives $\log Q_{m-1} \leq T^{1/2}$. The defining inequality for m and its fixed bound b_ν give $S_*^{m-1} \gg_\nu T^{(m-1)/(2m)} \geq T^{1/3}$. Thus $\log P_{m-1} \gg_\nu T^{1/3-11/10000}$, whereas $\log P_m \geq T^{1-22/10000}$ and $\log \log Q_m \leq \log T$. These bounds prove (8) at the last interval as well. For the first mean-value estimate in that proof, the case of X_1 on p. 36 of the arXiv version of [12], one needs $X/Q_1 \geq q\mathcal{T}$, where $\mathcal{T} = (X/H)R_*^{\rho/100}$ and $Q_1 = H/Q_* = R_*$. With $Q = q$, so that R_* would equal R , this inequality fails by the factor $R^{\rho/100}$; the printed condition $Q_1 \leq h/(QP_1^{0.01})$ is inconsistent with $Q_1 = h/Q$. The buffer in Q_* supplies the needed inequality:

$$q\mathcal{T} = \frac{X}{R} R_*^{\rho/100} \leq \frac{X}{R} R^{\rho/100} \leq \frac{X}{R_*} R^{-\rho/100} \leq \frac{X}{Q_1}.$$

The third inequality uses $Q_* \geq qR^{\rho/50}$. The external Q is a free parameter of Theorem 1.5 and Corollary 1.6, whose right-hand sides involve ε , $\varphi(q)$, X and H/q ; thus this buffer preserves their conclusions. The other uses of Q_* also hold: $R_*^{-0.01\rho} \ll \varepsilon$ and $q/H = 1/R \ll \varepsilon$ on p. 33 of the arXiv version, and $P_m \geq P_1$, since $\log P_1 \leq T^{2/5} \ll \rho^2 T = \log P_m$ on pp. 36–38 of that version.

The typicality in Definition 1.1 of [12] means that $\#\{p \mid q : p \leq t\} \leq \pi(t)/100$ for every $t \geq y$. Take $y = R_*^{\varepsilon^2}$. There are at most $\log X/\log 2$ distinct prime factors of q , whereas $\pi(y)/\log X \rightarrow \infty$: before capping, $\log y \asymp (\log R)^{0.998}$, and after capping $\log y \gg T^{0.398}$. All larger cutoffs satisfy the same condition. The cutoff $R_*^{\varepsilon^{1.1}}$ in Proposition 9.4 is larger than y .

Corollary 1.6, restricted to unit classes, now gives the variance about its real-character main term with error $O(\varepsilon\varphi(q)XR^2)$. A nonreal main term may be omitted in that corollary. For a real main term, Lemma 2.2 at cutoff $3X$ gives the relative squared cost

$$O((q/\varphi(q))^2(\log X)^{-1/20}) = o(\varepsilon),$$

using $q/\varphi(q) \ll \log \log(3q)$.

To include nonunits, group the classes by $u = (a, q)$ and put $n = um$, $x = ut$, $q' = q/u$, $X' = X/u$, $H' = H/u$. Then $H'/q' = R$, $R \leq X' \leq X$, $q' \leq X'$, and $dx = u dt$. At every such scale use the same $\varepsilon = (\log R)^{-1/1000}$. Choose its own Q_* by the displayed formula with X', H', q' . The zero-box and all lower-parameter checks are uniform for $X' \in [R, X]$: their worst lower scale is $X' = R$, where, for example, $\varepsilon^{-88} \log \log R/\log R = o(1)$. The typicality argument is repeated at that scale using $\omega(q') \leq \log X'/\log 2$ and the corresponding y ; one still has $\log R \geq (\log X')^\nu$. The function $(\log \log t)^2(\log t)^{-1/20}$ is decreasing for large t , so its value at X' is at most its value at R , which is $o(\varepsilon)$. Complete multiplicativity and $|\lambda(u)| = 1$ show that this gcd group contributes at most $C_\nu \varepsilon \varphi(q/u)XR^2$. Finally $\sum_{u|q} \varphi(q/u) = q$, which gives (6). $\square$

Lemma 2.4. Fix $0 < \nu_0 \leq 1$. For sufficiently large N and every real D with $\exp((\log N)^{\nu_0}) \leq D \leq N$, one has

$$\sup_{\theta \in \mathbb{R}} \frac{1}{N} \sum_{r=1}^N \left| \sum_{0 \leq k < \lfloor D \rfloor} \lambda(Nk + r) e(k\theta) \right| \ll_{\nu_0} D(\log D)^{-1/3000}. \quad (9)$$

Proof. Put $K_D = \lfloor D \rfloor$, $Q_0 = \lfloor \sqrt{D} \rfloor$. Dirichlet approximation gives $\theta = a/q_0 + \beta$ with $1 \leq q_0 \leq Q_0$ and $|\beta| \leq 1/(q_0 Q_0)$. Set

$$h = \lfloor \min\{D/(\log D)^{10}, q_0 Q_0/100\} \rfloor, \quad t_0 = h/(\log D)^{20}.$$

For $t_0 \leq t \leq h$, one has $t/q_0 \gg \sqrt{D}/(\log D)^{30}$, and $|\beta|h \leq 1/100$. For $0 \leq v < K_D$ define

$$U_{r,v}(t) = \sum_{1 \leq j \leq t} \lambda(N(v+j) + r) e(a(v+j)/q_0).$$

Group by $v + j \pmod{q_0}$ and apply Cauchy–Schwarz. The map $(r, b) \mapsto Nb + r$ is a bijection onto all classes modulo $Q = Nq_0$ for $1 \leq r \leq N$, $0 \leq b < q_0$. Moving the starting point $Nv + r$ to $Nv + u$, $0 \leq u < N$, changes each progression sum by at most two terms, since $Q \geq N$. Integrating in u gives

$$\sum_{r,v} |U_{r,v}(t)|^2 \ll \frac{q_0}{N} \int_0^{NK_D} \sum_{b \bmod Q} \left| \sum_{\substack{x < n \leq x+Nt \\ n \equiv b \pmod{Q}}} \lambda(n) \right|^2 dx + NK_D q_0^2. \quad (10)$$

For $0 \leq x < Nt$, each class has $O(t/q_0 + 1)$ terms, giving total cost $O(Nt^3)$ after multiplication by q_0/N . Cover the rest by dyadic intervals with starting scale $X' \geq Nt$; their starting scales have sum $O(ND)$. Lemma 2.3 applies with length Nt and modulus Q . One may take $\nu = \nu_0/2$, since for large N

$$(2 \log N + 1)^{\nu_0/2} \leq \frac{1}{2}(\log N)^{\nu_0} - 30 \log \log N - O(1) \leq \log(t/q_0).$$

Thus, with $\varepsilon_D = (\log D)^{-1/1000}$,

$$\frac{1}{NK_D} \sum_{r,v} |U_{r,v}(t)|^2 \ll_{\nu_0} \varepsilon_D t^2 + t^3/D + q_0^2 \ll_{\nu_0} \varepsilon_D t^2. \quad (11)$$

The final inequality uses $t/D \leq (\log D)^{-10}$ and $q_0/t \ll (\log D)^{30}/\sqrt{D}$.

Abel summation inserts $e(\beta j)$ into the window. At each $t \geq t_0$, average first in (r, v) and use (11); the mean absolute value is $O_{\nu_0}(\sqrt{\varepsilon_D} t)$. For $t < t_0$ use $|U_{r,v}(t)| \leq t$. The mean absolute value of the full θ window is consequently $O_{\nu_0}(h\sqrt{\varepsilon_D} + h(\log D)^{-40})$.

For any 1-bounded sequence (z_k) the deterministic identity

$$\frac{1}{h} \sum_{v=0}^{K_D-1} \sum_{j=1}^h z_{v+j} = \sum_{k=0}^{K_D-1} z_k + O(h+1)$$

follows by counting the multiplicity of each index: the sum of the absolute differences of the coefficients is at most $h+1$. Use it with $z_k = \lambda(Nk+r)e(k\theta)$ and average in r . The resulting bound is $O_{\nu_0}(D\sqrt{\varepsilon_D} + h+1)$, which is $O_{\nu_0}(D(\log D)^{-1/2000})$ and implies (9). The frequency is fixed throughout each average, so the supremum remains outside the r average. $\square$

3 Finite graph inputs and their parameters

Throughout this section the prefix of a family 007 Lean name is `OAI.TwoPointCorrelations`. Thus the names printed below specify fully qualified declarations in that namespace.

Definition 3.1 (Arithmetic word data). A prime family consists of disjoint finite prime sets $\mathcal{P}, \mathcal{Q}$ and a finite supply $\mathcal{S}$ of pairs (d, q) , where d, q are squarefree, d has J prime factors in $\mathcal{P}$, and q has at most m_Q prime factors in $\mathcal{Q}$. At multiplier $h \geq 1$, every $p \in \mathcal{P}$ is required not to divide h . This is the mathematical data of `ProhibitedPrimeFamily`. For a signed step $a = (\epsilon, d, q)$, $\epsilon \in \{-1, 1\}$, put $\Delta_h(a) = \epsilon h q d$ and $\Delta_h(w) = \sum_{a \in w} \Delta_h(a)$ for a word w . A word is positive at $x \in \mathbb{Z}$ when, for each successive step, its whole divisor $q d$ divides the departure integer $x + \Delta_h(w_{<i})$.

For completeness we specify the prohibited predicate used in the graph. A word $w = (a_0, \dots, a_{R-1})$ is forward-prohibited if $3 \leq R \leq s$, all its pairs belong to $\mathcal{S}$, consecutive tuple labels differ, and each tuple prime occurs on an interval of indices. There must also be a prime p in the first tuple but not in the last, and an index $0 < a < R-1$, such that $p \mid \Delta_h(w_{\geq a})$. A

forward-prohibited word is minimal if none of its shorter contiguous subwords, in either orientation, is forward-prohibited. Reversal changes the order of the steps and reverses every direction. An integer is prohibited if some minimal forward-prohibited word is positive there. This is exactly `ProhibitedSite`, built from `PositiveWord`, `ForwardProhibited`, and `MinimalWord`. Positivity here is a divisibility condition for integers of either sign.

The residue model assigns independent uniform variables $R_p \in \mathbb{Z}/p\mathbb{Z}$ for $p \in \mathcal{P} \cup \mathcal{Q}$. Integer divisibility at $x + u$ is replaced by $R_p + x + u = 0 \pmod{p}$. A common CRT lift of these variables defines every word predicate; the choice of lift has no effect. All expectations in the remainder of this section are with respect to this finite probability space.

Input 3.2 (Closed-word column bound). *Fix $h \geq 1$, $C \geq 0$ and $W \geq 1$. For all sufficiently large L , the following holds for every prime family of Definition 3.1. Suppose its center set is the union of pairwise disjoint bands $\mathcal{P}_1, \dots, \mathcal{P}_J$, with masses $V_i = \sum_{p \in \mathcal{P}_i} p^{-1}$, and suppose*

$$J \geq 1, \quad J + m_Q \leq C \log L, \quad m_Q \leq 100 \log L, \quad 1 \leq V_i \leq \min(2W, L^2), \quad (12)$$

$$1 \leq \sum_{p \in \mathcal{P}} p^{-1} \leq L^2, \quad \sum_{p \in \mathcal{Q}} p^{-1} \leq L^2, \quad (13)$$

$$\exp(L^{199/200}) \leq H \leq p \leq Y \leq \exp L \quad (p \in \mathcal{P}), \quad p \leq B \leq \exp L \quad (p \in \mathcal{P} \cup \mathcal{Q}), \quad (14)$$

$$q \leq Q_{\max} \leq \exp(100L + 1) \quad ((d, q) \in \mathcal{S}), \quad \prod_{i \neq j} p_i \leq D_{\max} \leq \exp(2L) \quad (p_i \in \mathcal{P}_i). \quad (15)$$

Set $k = \lfloor L \rfloor$, $s = \lfloor L^{1/10} \rfloor$, and $D_b = 2k + \lfloor L^{1/12} \rfloor s$. Let $\mathcal{D}$ be the tuple products, and take any finite squarefree padding-divisor set supported on $\mathcal{Q}$. Use a nonnegative padding coefficient $u(q) \leq 4^{\omega(q)}$, a positive Q -residue-dependent vertex weight g with $g^2 \geq 1$, a row cap $K \geq 1$, and Q -residue-dependent extra cuts. Retain precisely the eligible edges, the row-density cap, these extra cuts, and the nonprohibited vertices. Center each tuple by $\prod_{p|d} (\mathbf{1}_{p|x} - 1/p)$.

For any finite state set mapped into tuple labels times integer sites, any deterministic edge gates, any fixed initial state and any prescribed sequence of $2k$ directions, let $\mathcal{F}$ be the numerical closed-word catalog obtained from pairs of k -step paths with distinct consecutive tuple labels in each half. No extra condition is imposed at the seams. If Z_w is the actual signed masked scalar word weight, then

$$\exp(108L) \sum_{w \in \mathcal{F}} |\mathbb{E} Z_w| \leq \{K(2 \exp(150) \sqrt{W})^J\}^{2k}. \quad (16)$$

The state-to-site map need not be injective in this column statement.

The exact name is `eventually_actual_column_trace`. The numerical catalog is its `actual ClosedPairCatalog`, with `closedTraceFiber` fixing the direction sequence. The theorem takes the threshold in L after h, C, W are fixed and before all prime families, numerical words, gates and state maps. The more general theorem `eventually_prohibited_column_trace_total` is the source of this column bound. The correspondence between the signed masked word and the centered residue expectation is `maskedClosedWord_residue_average`: for a closed word of positive length $R \leq D_b$, the residue average of the actual masked word equals the centered product times its attached-catalog avoidance factor and retained departure weight. Its hypotheses are disjoint prime bands, the matching label assignment and a nowhere-zero g . On a closed walk the square-root vertex denominators combine into the departure squares; the idempotent vertex masks combine into the attached avoidance condition. This equality retains the signs before taking the absolute values in (16).

Input 3.3 (Finite comparisons). *There is an absolute integer $A_f \geq 1000$ such that, for all sufficiently large L , the following comparisons are uniform after L is fixed. The coordinate moduli are positive and pairwise coprime, number at most $\exp L + 1$, and are at most $\exp L$. A Boolean test can be any unary test of a specified residue coordinate. A depth-at-most-20 circuit of size at most $\exp(L^6)$ has empirical versus independent-residue error at most $\exp(-L^{10})$ on any integer interval of length $U \geq \exp(L^{A_f}/2)$.*

In the following actual-word and positive-cost comparisons we use $u(q) = 4^{\omega(q)}$ and $g(n)^2 = 5^{\omega_Q(n)}$, with the Q -degree cut $\omega_Q(n) \leq 400 \log L$. These comparisons concern these arithmetic weights and the positive deletion atoms defined in Section 4. For an actual closed masked word the error is at most $\exp(-L^9)$. Its word length R satisfies $0 < R$, $R + 1 \leq 4L$, $RJ \leq L^2$; $s \leq L$, $J + m_Q \leq L^2$, the supply has size at most $\exp(101L)$, and every padding choice has at most $100 \log L$ prime factors. The Q -degree cut is at $\lfloor 400 \log L \rfloor$. The comparison holds on $a + \ell\{0, \dots, U - 1\}$ for arbitrary $a, \ell \geq 0$ with $(\ell, p) = 1$ for every pool prime.

Positive deletion atoms satisfy an error of at most $2\exp(-L^9)$, uniformly in the prime family and any integer offset. The average degree-truncated padding weight is at most its product-law normalizer plus $\exp(-L^9)$, and at most twice that normalizer. Sums of atoms incur the sum of the individual absolute errors. Bin-dependent interval lengths may be used, provided each is at least $\exp(L^{A_f}/2)$.

The first statement is `BravermanDepth22Input.eventually_residue_comparison`, obtained from `BravermanDepth22Input.finite_residue_comparison`. The actual-word statement is `BravermanDepth22Input.eventually_actual_affine_word_comparison`. The last statements are the specializations of

```
BravermanDepth22Input.eventually_variable_padding_comparison,
BravermanDepth22Input.eventually_prohibited_row_comparison,
BravermanDepth22Input.eventually_variable_tuple_degree_deletion,
BravermanDepth22Input.eventually_actual_padding_weight.
```

The first, unweighted comparison is stated for consecutive origins. Its arbitrary unary tests can be precomposed with $z \mapsto a + \ell z$; multiplication by an invertible ℓ preserves the independent uniform law. The named affine word theorem performs this precomposition for the full weighted word. There is no upper bound on a, ℓ , nor a requirement that the product of all pool primes be small. The proved premise `bravermanDepth22Input` supplies the circuit comparison.

The weighted comparison expands at most $4L$ active-state slots, each with at most $\exp L$ coordinates and at most $400 \log L$ active bits, and at most L^2 center bits. The coefficient table is bounded by $\exp(L^4)$, and the surviving-site circuit has depth at most 19 and size at most $\exp(L^3)$. This is the exact budget in `BravermanDepth22Input.eventually_affine_weighted_word_comparison`. It gives the actual-word and positive-atom versions above. Arbitrary offsets change the unary residue tests and preserve these budgets.

Input 3.4 (Prohibited sites and spectral transfer). *For every fixed $C \geq 0$, for all sufficiently large L , uniformly in the multiplier h , the prime family and its supply,*

$$\mathbb{P}(\text{a residue-model site is prohibited}) \leq \exp(-\tfrac{1}{2}L^{199/200}) \quad (17)$$

if $1 \leq s \leq L^{1/10}$, $J + m_Q \leq C \log L$, the pool masses are at most L^2 , the center mass is at least 1, and all center primes are at least $\exp(L^{199/200})$, with all pool primes at most $\exp L$.

For any finite-dimensional complex Hilbert space, finite self-adjoint family (B_d) , orthogonal projection E , and nonnegative real numbers a, b , suppose

$$\|B_d\| \leq a, \quad \sum_d \|B_d v\|^2 \leq b^2 \|v\|^2 \quad (Ev = v).$$

If $\mathcal{H}$ is its nonbacktracking operator, then

$$\left\| E \sum_d B_d E \right\| \leq 3 \max\{a, b, \text{rad}(\mathcal{H})\}. \quad (18)$$

The exact names are `eventually_prohibited_density` and `noncommuting_spectral_transfer_max`; the common-bound version is `noncommuting_spectral_transfer`. The probability bound follows from the finite theorem `ProhibitedPrimeFamily.probability_bound`, which includes the whole prohibited catalog. This probability bound concerns a single site. In Lemma 4.4 it is combined by Cauchy–Schwarz with second moments of the positive weights, and sums over many sites are compared with the residue model by Input 3.3. The nonbacktracking operator uses distinct successive tuple labels; different B_d need not commute, and E need not commute with them.

Lemma 3.5 (A numerical finite-comparison exponent). *Every finite comparison used here may be run with $A_f = 1000000$.*

Proof. We record the exponent extraction from the explicit circuit budgets in [23]. Write $\mathcal{G}(f, c)$ for $\exists K \forall j \geq 0 : f(j) \leq K(j+1)^c$. The proof of `bravermanDegree_polynomial` uses the rules

$$\mathcal{G}(f, c), \mathcal{G}(g, d) \implies \mathcal{G}(f + g, c + d), \quad \mathcal{G}(fg, c + d), \quad \mathcal{G}(f^r, rc).$$

Constants have exponent 0 and j has exponent 1. Applying these rules in the order of the source definitions gives

Budget	Polynomial exponent in $j + 1$
<code>bravermanBase</code>	2
<code>bravermanNormExponent</code>	45
<code>bravermanErrorExponent</code>	1
<code>bravermanSwitches</code>	47
<code>switchingDenominator</code> at that switch count	94
<code>switchingDegree</code> at depth 89	$47 + 89 \cdot 94 = 8413$
<code>bravermanDegree</code>	$44 + 8413 = 8457$

Increasing the positive-growth exponent by one, as in `bravermanDegree_polynomial_positive`, gives a fixed bound $K(j+1)^{8458}$ for some absolute $K > 0$. The theorem `ReflectedLiouville.braverman_degree_positive_explicit_exponent` in the ancillary file `ExplicitBudget.lean` proves this numerical exponent for the source’s literal degree function. The dyadic comparison `AC0Circuit.dyadic_comparison` requires independence degree at least this explicit `bravermanDegree`. The logarithmic-budget conversion changes K , not 8458.

In the finite-residue comparison take bit depth $b_L = \lceil L^{15} \rceil$, accuracy $\varepsilon_L = \exp(-L^{11})$, and independence order $t_L = \lceil L^{17 \cdot 8458 + 1} \rceil = \lceil L^{143787} \rceil$. The encoded size is at most $\exp(L^{17})$; hence, for large L , its required independence order is at most t_L . The total number of encoded bits is at most $\exp(2L)$. The low-order CRT/Fourier error is bounded by

$$\frac{(\exp L)^{t_L}}{U} (t_L + 1) (\exp(2L))^{t_L} \leq \exp(-L^{11}) \quad \text{if } U \geq \exp(L^{1000000}/2).$$

The two decoder errors total at most $2 \exp(3L) 2^{-b_L}$, so the finite formula gives error at most $\exp(-L^{10})$. The signed-state expansions and all positive costs have total coefficient mass at most $\exp(CL^5)$ for a fixed C ; this changes the error to at most $\exp(-L^9)$ and does not change the required

interval exponent. In the source, `BravermanDepth22Input.eventually_residue_comparison` takes $T := 17C + 1$ and $A := T + 1002$, where C is the exponent of `bravermanDepth22Input`. Here $C = 8458$, so $A = 17 \cdot 8458 + 1003 = 144789$. The comparisons `eventually_scalar_residue_comparison`, `eventually_active_state_comparison`, `eventually_weighted_word_comparison`, `eventually_affine_weighted_word_comparison`, `eventually_actual_affine_word_comparison`, `eventually_positive_weight_comparison`, `eventually_positive_prime_cost_comparison`, `eventually_prohibited_cost_comparison`, `eventually_prohibited_row_comparison`, `eventually_degree_cost_comparison`, `eventually_integer_degree_cost`, `eventually_integer_degree_sum`, `eventually_variable_tuple_degree_deletion`, `eventually_state_cost_comparison`, `eventually_density_cost_comparison`, via the state-cost comparison, and `eventually_actual_padding_weight`, the last via `eventually_prime_state_weight`, each retain the exponent of their predecessor. The theorem `eventually_padding_atom_comparison`, whose exponents A_1, A_2 come from `eventually_degree_cost_comparison` and `eventually_density_cost_comparison`, and hence `eventually_variable_padding_comparison`, takes $A_1 + A_2 = 2 \cdot 144789 = 289578$. All are at most 10^6 . Enlarging A strengthens the hypothesis $U \geq \exp(L^A/2)$, proving the stated common choice. $\square$

4 Positive budgets and a rough-number multiplier

Fix, for the rest of Part I,

$$A = 1000000, \quad W = 10^{180}, \quad \delta = 10^{-5}, \quad \gamma = 1 - 2\delta, \quad s_0 = 10^{-5}. \quad (19)$$

These constants precede every sufficiently-large threshold. For an integer N put

$$L = (\log N)^{1/A}, \quad J = \left\lfloor \frac{\delta \log L}{6W} \right\rfloor, \quad \eta = \exp(-J), \quad K = \exp(4J), \quad Y_i = L^{1-\delta} \exp(6Wi). \quad (20)$$

We henceforth take N large enough that $J \geq 1$. For $1 \leq i \leq J$ let

$$\mathcal{P}_i = \{p : p \text{ prime, } p \equiv 1 \pmod{5}, p \nmid N, Y_{i-1} \leq \log p < Y_i\}, \quad \mathcal{P} = \bigcup_i \mathcal{P}_i,$$

and put

$$H_0 = \exp(L^{1-\delta}), \quad \mathcal{Q}_0 = \{p \leq \exp L : p \text{ prime, } p \not\equiv 1 \pmod{5}\}, \quad \mathcal{Q} = \{p \in \mathcal{Q}_0 : p \nmid N\}.$$

The fixed-modulus prime number theorem and partial summation give, uniformly in i , the full band mass $3W/2 + o(1)$. Deleting prime factors of N from all center bands costs at most

$$\sum_{p|N, p \geq H_0} \frac{1}{p} \leq \frac{\log N}{H_0 \log H_0} = o(1). \quad (21)$$

Thus $W \leq V_i \leq 2W$ for all i once N is large. The estimate is uniform in N , because $\log N = L^A$ and $H_0 = \exp(L^{1-\delta})$.

Let $\mathcal{D}$ consist of products $d = \prod_{i=1}^J p_i$, $p_i \in \mathcal{P}_i$, and let $V_* = \prod_i V_i$. The bands are disjoint, so the tuple is determined by d , and

$$\log d < \sum_{i=1}^J Y_i < 2L, \quad \sum_{d \in \mathcal{D}} \frac{1}{d} = V_*. \quad (22)$$

Every pool prime is at most $\exp L$. For squarefree products q of primes in $\mathcal{Q}$ put

$$u(q) = 4^{\omega(q)}, \quad w(x) = 5^{\omega_{\mathcal{Q}}(x)}, \quad S = \prod_{p \in \mathcal{Q}} (1 + 4/p).$$

For arbitrary integers x , $\omega_{\mathcal{Q}}(x)$ counts divisibility by this finite prime set. In particular this definition also applies at negative integers and at zero.

Retain pairs (d, q) with $\omega(q) \leq 100 \log L$ and $\log(qd) \leq 100L + \eta$. Define disjoint bins

$$\mathcal{S}_j = \{(d, q) \text{ retained} : j\eta \leq \log(qd) < (j+1)\eta\}, \quad D_j = \exp(j\eta), \quad T_j = ND_j. \quad (23)$$

Only nonempty bins are used; their number is $O(L \exp J)$. The union $\mathcal{S}$ of all bins is the word supply in Definition 3.1. For each d the retained divisor set is denoted by $\mathcal{Q}_d$. The whole-product cutoff is included in $\mathcal{Q}_d$ before binning.

Lemma 4.1 (Retained mass). *For all sufficiently large N ,*

$$S_0 := \sum_j \sum_{(d,q) \in \mathcal{S}_j} \frac{u(q)}{qd} \geq \frac{1}{2} S V_*. \quad (24)$$

Moreover, $\#\mathcal{S} < \exp(101L)$ and $m_{\mathcal{Q}} := \lfloor 100 \log L \rfloor$ is a valid padding-factor cap.

Proof. The probability law proportional to $u(q)/q$ independently selects each padding prime with probability $4/(p+4)$. Its mean $\log q$ is at most $4L + O(1)$, by $\sum_{p \leq \exp L} \log p/p = L + O(1)$. For every d , (22) and Markov give $\mathbb{P}(\log(qd) > 100L + \eta) < 1/10$ for large L . Also

$$\mathbb{E} \exp(\omega(q)) \leq \exp(4(\exp(1) - 1) \sum_{p \leq \exp L} 1/p) \ll L^{4(\exp(1)-1)},$$

so $\mathbb{P}(\omega(q) > 100 \log L) = o(1)$. At least half of the padding mass is retained for each d ; summing $1/d$ proves (24). These upper-tail estimates remain valid after deleting any prime factors of N . Finally, dq determines its P-part d and Q-part q , and every retained product is at most $\exp(100L + \eta) < \exp(101L)$. $\square$

For $d \in \mathcal{D}$ define the positive center envelope

$$a_d^+(x) = \prod_{p|d} (1_{p|x} + 1/p), \quad \rho_{j,d}(x) = \frac{1}{w(x)} \sum_{q: (d,q) \in \mathcal{S}_j} u(q) \mathbf{1}_{q|x}.$$

The densities sum to at most 1 over all bins, since $\sum_{q|x, q \text{ squarefree on } \mathcal{Q}} u(q) = w(x)$. Let

$$F_d(x) = w(x) \sum_j \rho_{j,d}(x) \mathbf{1}_{\{\rho_{j,d}(x) > K/L \text{ or } \omega_{\mathcal{Q}}(x) > 400 \log L\}}.$$

The corresponding definitions for $\mathcal{Q}_0$ use the same $\mathcal{Q}_d$ and bins, replacing only the vertex weight and degree by their full-pool versions.

Input 4.2 (Full-pool padding cost). *There is an absolute $C_p > 0$ such that, for all sufficiently large L , for every prime family with Q-pool $\mathcal{Q}_0$, every retained squarefree divisor subset, every finite bin set, every $0 < \eta \leq 1$, every $K > 0$, and every integer offset,*

$$\frac{\mathbb{E}\{a_d^+(x) F_{d,0}(x)\}}{S_{\text{full}}} \leq \frac{2^J}{d} \{C_p/K + L^{-100}\}, \quad S_{\text{full}} = \prod_{p \in \mathcal{Q}_0} (1 + 4/p). \quad (25)$$

The center primes in d belong to the disjoint P-pool.

This is the squarefree-tuple specialization of `ModFiveThetaInput.eventually_positive_padding_cut_cost`, with its fixed excluded set $E = \emptyset$; the proved input is `modFiveThetaInput`. The hypothesis on its divisor subset is just containment in the squarefree products of the Q-pool. In particular it permits $\mathcal{Q}_d$ rather than a complete divisor set. In the source the excluded set is fixed before the threshold in L ; all the displayed prime families, subsets, bins and offsets follow that threshold.

Lemma 4.3 (Padding holes). *Uniformly in N and in the bins and retained sets just defined,*

$$\frac{1}{SV_*} \sum_{d \in \mathcal{D}} \mathbb{E}\{a_d^+(x) F_d(x)\} \ll 2^J \{K^{-1}(\log L)^{10} + L^{-90}\}. \quad (26)$$

Proof. By translation invariance of the independent uniform residue law, take the single site to be zero. At a fixed site x , the equivalent condition is $R_p + x \not\equiv 0 \pmod{p}$ for every removed prime. Put $E = \mathcal{Q}_0 \setminus \mathcal{Q}$ and let $A_E = \{R_p \neq 0 \text{ for every } p \in E\}$ in the full product model. Then $\mathbb{P}(A_E) = \prod_{p \in E} (1 - 1/p)$. Conditioned on A_E , the remaining coordinates have exactly the reduced-pool law; at this one site the full and reduced weights, degrees and retained-bin densities agree. The retained set $\mathcal{Q}_d$ is identical on the two sides. The center envelope uses disjoint coordinates. Nonnegativity gives

$$\frac{\mathbb{E}\{a_d^+ F_d\}}{S} \leq \frac{\mathbb{E}\{a_d^+ F_{d,0}\}}{S_{\text{full}}} \prod_{p \in E} \frac{p+4}{p-1}.$$

The sum of reciprocals of the prime divisors of N is at most $\log \log N + O(1)$: use Mertens up to $\log N$ and the bound $\omega(N) \leq \log N / \log 2$ above it. Consequently the last product is $O((\log \log N)^{10}) = O_A((\log L)^{10})$. Apply (25) and sum $1/d = V_*$. The extra logarithmic factor on L^{-100} is absorbed by L^{-90} . Only single-site padding costs have been conditioned in this argument. $\square$

Lemma 4.4 (Center and prohibited costs). *Uniformly in all the parameters above and in any residue translation, the positive center-degree cost, summed over all bins, is at most*

$$S 2^J V_* \exp(-2WJ) \quad (27)$$

when vertices with $\omega_{\mathcal{P}}(x) > 6WJ$ are deleted. The positive prohibited-site cost is at most

$$SV_* \exp(-L^{0.9}). \quad (28)$$

These costs have the same bounds for integer intervals of the bin lengths, up to an error $\exp(-\frac{1}{2}L^9)$.

Proof. For a fixed d , the mean of a_d^+ is $2^J/d$. Tilting by this envelope makes a selected tuple prime active with probability $(p+1)/(2p)$; all other center primes retain probability $1/p$. The coordinates remain independent. Their exponential moment is bounded by $\exp((\exp(1) - 1)(2W + 1)J)$. Since $W \geq 10$,

$$\mathbb{P}_{a_d^+}(\omega_{\mathcal{P}} > 6WJ) \leq \exp(((\exp(1) - 1)(2W + 1) - 6W)J) \leq \exp(-2WJ).$$

Multiply by $2^J/d$, sum over d , and use independence of the Q-pool and $\sum_j \rho_{j,d} \leq 1$. This gives (27).

For prohibited sites the positive row envelope is

$$A_+(x) = w(x) \prod_{i=1}^J (\omega_{\mathcal{P}_i}(x) + V_i).$$

Mertens and independence give

$$\mathbb{E}w^2 = \prod_{p \in \mathcal{Q}} (1 + 24/p) \ll L^{24}, \quad \mathbb{E}A_+^2 \leq CL^{24} \prod_i (V_i + 4V_i^2) \leq CL^{24} 5^J V_*^2.$$

The prohibited probability is bounded by (17), with $h = N$, $s = \lfloor L^{1/10} \rfloor$, and the full retained supply $\mathcal{S}$. Its hypotheses hold because $J + m_Q \leq 101 \log L$, both pool masses are at most L^2 , and $H_0 \geq \exp(L^{199/200})$. Cauchy–Schwarz bounds the positive rare-row mean by $CL^{12} 5^{J/2} V_* \exp(-\frac{1}{4} L^{199/200})$. Since $J = O(\log L)$ and $S \geq 1$, this is at most the right side of (28), even after multiplying by the number of bins.

For the integer statements use the positive-atom comparisons in Input 3.3, individually before summing. There are at most $\exp(O(L))$ atoms and polynomially many bins. The selected-prime count is at most L^2 , each selected padding divisor has at most $100 \log L$ prime factors, each whole divisor has at most $J + 100 \log L \leq L^2$ prime factors, the supply has at most $\exp(101L)$ pairs, $W \geq 10$ and $6WJ \leq \delta \log L \leq 400 \log L$. These are the hypotheses of `eventually_prohibited_row_comparison`, `eventually_variable_tuple_degree_deletion` and `eventually_variable_padding_comparison`. Every interval length $\lfloor T_j \rfloor$ is at least N , and hence at least $\exp(L^A/2)$. The total comparison error is thus $\exp(-\frac{1}{2} L^9)$ for large L . Arbitrary offsets are included in the comparison. Product-law translation invariance supplies the target-site versions. $\square$

Lemma 4.5 (Rough multiplier). *Fix $0 < \gamma_1 < \kappa_1 < 1$. For all sufficiently large L , let $y = \exp(L^{\gamma_1})$ and $D \geq \frac{1}{2} \exp(L^{\kappa_1})$. If $Z \subset [D, 2D) \cap \mathbb{Z}$ consists of y -rough integers and $|c_n| \leq 1$, then*

$$\left\| \sum_{n \in Z} \frac{c_n}{n} e(n\theta) \right\|_{\infty} \ll L^{-\gamma_1}, \quad \int_0^1 \left| \sum_{n \in Z} \frac{c_n}{n} e(n\theta) \right|^4 d\theta \ll D^{-1} L^{-4\gamma_1}. \quad (29)$$

The constants may be taken absolute; the threshold may depend on γ_1, κ_1 .

Proof. We bound both one rough integer and the four forms $a, b, c, a+b-c$ in the three-dimensional box $[D, 2D)^3$. For a prime p their simultaneous nonvanishing density is exactly

$$\frac{(p-1)^3 - (p-1)(p-2)}{p^3} = 1 - 4/p + 6/p^2 - 3/p^3.$$

Indeed there are $(p-1)(p-2)$ nonzero pairs a, b with $a+b \neq 0$, each excluding one nonzero value $c = a+b$. The bad-event density ν_p is at most $4/p$. Take the even Bonferroni order $r_0 = 2 \lceil 500 \log L \rceil$. The independent truncation error is at most

$$\frac{(\sum_{p \leq y} \nu_p)^{r_0+1}}{(r_0+1)!} \leq \frac{(5 \log L)^{r_0+1}}{(r_0+1)!} \ll L^{-100}.$$

Each truncated subset has CRT modulus $q \leq y^{r_0}$. The count in a fixed residue triple is $(D/q)^3 + O((D/q)^2 + D/q + 1)$. Summing over its at most q^3 residue triples gives normalized error $O(q/D)$, since $q \leq D$ for large L . There are at most $(r_0+1)(y+1)^{r_0}$ subsets, so the total error is

$$D^{-1} \exp(O(L^{\gamma_1} \log L)) \ll L^{-100}.$$

The one-dimensional CRT estimate has the same or smaller error. Mertens' theorem bounds the independent good densities by $CL^{-\gamma_1}$ and $CL^{-4\gamma_1}$, respectively; the ratio of the four-form product to $\prod_{p \leq y} (1 - 1/p)^4$ is bounded because the remaining logarithmic terms are $O(p^{-2})$. Consequently

$$|Z| \ll DL^{-\gamma_1}, \quad \#\{a, b, c, d \in Z : a+b = c+d\} \ll D^3 L^{-4\gamma_1}.$$

The first inequality proves the sup norm bound. Expanding the fourth moment and integrating leaves precisely the additive quadruples; each coefficient has absolute value at most D^{-4} . The second inequality gives the remaining assertion. $\square$

5 The reflection graph and proof of the main theorem

Extend λ by zero to nonpositive integers only in the test functions of this section. For a bin set

$$R_j = \frac{1}{T_j} \sum_{1 \leq x \leq T_j} \sum_{(d,q) \in \mathcal{S}_j} u(q) \mathbf{1}_{qd|x} \lambda(x) \lambda(Nqd - x), \quad (30)$$

$$C_j = \frac{1}{T_j} \sum_{1 \leq x \leq T_j} \sum_{(d,q) \in \mathcal{S}_j} u(q) \mathbf{1}_{q|x} \prod_{p|d} (\mathbf{1}_{p|x} - 1/p) \lambda(x) \lambda(Nqd - x). \quad (31)$$

Every real cutoff in these sums means its integer floor; the displayed normalizing denominator remains the real number T_j .

Lemma 5.1 (Dilation identity).

$$\sum_j R_j = S_0 \frac{C(N)}{N} + O(SV_*(\eta + N^{-1})). \quad (32)$$

Proof. For each pair in (30), put $x = qdm$. Complete multiplicativity cancels $\lambda(qd)^2 = 1$, and the cutoff $T_j/(qd)$ lies in $(N \exp(-\eta), N]$. The lost tail of a 1-bounded sum is at most $N\eta + 1$. Also $T_j/(Nqd) \in (\exp(-\eta), 1]$. Thus that summand equals $u(q)(qd)^{-1} C(N)/N + O(u(q)(qd)^{-1}(\eta + N^{-1}))$. Summing proves the assertion, since $S_0 \leq SV_*$. No total N has been averaged. $\square$

Lemma 5.2 (CRT transfer). *Let $P_* = \prod_{p \in \mathcal{P} \cup \mathcal{Q}} p$. Choose a positive integer ℓ with $N\ell \equiv 1 \pmod{P_*}$. For every $r, k, \xi \in \mathbb{Z}$ and every pool prime,*

$$p \mid Nk + r + N\xi \iff p \mid k + \ell r + \xi. \quad (33)$$

All word divisibility, centered factors, Q -weights and Q -degree tests therefore transfer from physical multiplier N to quotient multiplier 1. The full prohibited predicate satisfies

$$\text{Prohibited}_N(Nk + r) \iff \text{Prohibited}_1(k + \ell r). \quad (34)$$

Proof. Every pool prime is coprime to N , and multiplication by ℓ proves (33). The squarefree whole divisors qd consequently have the same divisibility at every departure point of a numerical word. Its displacements scale by N . The suffix condition in a forward-prohibited word becomes $p \mid N\Delta_1(w_{\geq a}) \iff p \mid \Delta_1(w_{\geq a})$, since its p belongs to the center pool. The tuple intervals, unequal successive labels, length bounds and supply membership do not change. Hence the forward-prohibited condition agrees for every word, as does its minimality over oriented contiguous subwords. Together with positivity this proves (34). $\square$

The corresponding finite identity is also supplied by `ReflectedLiouville.prohibited_CRT` in the ancillary file `CRTTransport.lean`; its hypotheses are precisely the coprimality of every qd with N and the displayed inverses at pool primes. The Liouville arguments are still the physical integers.

Fix a bin and write each integer uniquely as $x = Nk + r$, $1 \leq r \leq N$. Let $M_B = \lceil \exp(103L) \rceil$ and take two copies of $[-M_B, M_B] \cap \mathbb{Z}$. For every $d \in \mathcal{D}$ connect the left k to the right $t = k - qd$ whenever $(d, q) \in \mathcal{S}_j$, and include the transpose edge. Put

$$c_d(x) = \prod_{p|d} (1_{p|x} - 1/p), \quad \tau_{j,d}(x) = \mathbf{1}_{\{\rho_{j,d}(x) \leq K/L, \omega_{\mathcal{Q}}(x) \leq 400 \log L, x \text{ nonprohibited at multiplier } N\}}.$$

The left-to-right entry of the resulting self-adjoint matrix is

$$(B_{d,r})_{k,t} = \frac{Lu(q)\tau_{j,d}(Nk+r)\tau_{j,d}(Nt+r)}{\sqrt{w(Nk+r)w(Nt+r)}} \mathbf{1}_{q|Nk+r} c_d(Nk+r), \quad t = k - qd. \quad (35)$$

Every other left-to-right entry is zero. Let E_r be the diagonal projection onto center degree at most $6WJ$ in both layers. Its values use the physical sites $Nk + r$, including negative sites.

Proposition 5.3 (Compression). *With the absolute choice $C_3 = 3^{170}$, for all sufficiently large N , all but at most $N \exp(-2\lfloor L \rfloor)$ values of r satisfy*

$$\left\| E_r \sum_{d \in \mathcal{D}} B_{d,r} E_r \right\| \leq K(C_3 \sqrt{W})^J. \quad (36)$$

The exceptional set may depend on the bin.

Proof. Let $g_r(k) = \sqrt{w(Nk+r)}$ and $a_d(k) = |c_d(Nk+r)|$. For a fixed label, different q give different neighbors in each layer. The center factor and the test $q \mid Nk+r$ agree at the two endpoints because qd divides their physical difference Nqd . From the density cap,

$$\sum_t |(B_{d,r})_{k,t}| \frac{g_r(t)}{g_r(k)} \leq La_d(k) \rho_{j,d}(Nk+r) \mathbf{1}_{\{\rho_{j,d}(Nk+r) \leq K/L\}} \leq Ka_d(k).$$

The same calculation holds for a row in the right layer. Weighted Cauchy–Schwarz and self-adjointness give

$$\|B_{d,r}v\|^2 \leq K^2 \sum_k a_d(k)^2 |v(k)|^2, \quad \|B_{d,r}\| \leq K. \quad (37)$$

To see the first inequality, bound each row square by $Ka_d(k) \sum_t |B_{k,t}| g_r(k) g_r(t)^{-1} |v(t)|^2$, sum in k , and use $a_d(k) = a_d(t)$ on every nonzero entry. The finite lemma `ReflectedLiouville.two_1_ayer_localized_square` in the ancillary file `TwoLayerRows.lean` has exactly this weighted-row specialization and permits the duplicated site projection. On $E_r v = v$, the factorization over the disjoint bands gives

$$\sum_d a_d(k)^2 \leq \prod_i (\omega_{\mathcal{P}_i}(Nk+r) + V_i) \leq (8W)^J.$$

The last inequality is AM–GM, because the sum of these J factors is at most $6WJ + 2WJ$. Thus (18) applies to the self-adjoint family $(B_{d,r})_{d \in \mathcal{D}}$ and the projection E_r with the nonnegative reals $a = K$ and $b = K(8W)^{J/2}$, by (37) and the last display.

Let $\mathcal{H}_r$ be the nonbacktracking operator on the direct sum of one copy of the two-layer space for each d : $(\mathcal{H}_r v)_d = \sum_{e \neq d} B_{e,r} v_e$. All $B_{e,r}$ are real symmetric, so $(\mathcal{H}_r^* v)_d = B_{d,r} \sum_{e \neq d} v_e$, $\|\mathcal{H}_r^k\|_{\text{HS}} = \|(\mathcal{H}_r^*)^k\|_{\text{HS}}$ and $\text{rad}(\mathcal{H}_r^*) = \text{rad}(\mathcal{H}_r)$. Set $k_* = \lfloor L \rfloor$ and $B_0 = K(2 \exp(150) \sqrt{W})^J$, and expand $\|(\mathcal{H}_r^*)^{k_*}\|_{\text{HS}}^2$. At each step the factor is $B_{d,r}$, where d is the tag of the arrival state. This is the convention of the source’s `shiftMatrix` and `integerShiftNext`: the new state is the pair consisting

of the step tag and the shifted site. An entry from (e_0, x_0) to (d_{k_*}, x_{k_*}) sums paths with labels $d_1, \dots, d_{k_*}$ satisfying $d_1 \neq e_0$ and $d_i \neq d_{i+1}$. The squared Hilbert–Schmidt norm sums pairs of such paths with the same initial and terminal states. The initial tag e_0 is the fixed root tag, counted among the $\leq \exp(106L)$ states below; the common terminal tag is the common last step label $d_{k_*} = d'_{k_*}$. Reversing the second path gives a closed numerical word of length $2k_*$, nonbacktracking in each half, with unconstrained seams. The pair belongs to `actualClosedPairCatalog` for (e_0, x_0) , since `closedPairNumericalTest` compares the full end states: last label and integer site. For each numerical word and initial layer the lifted path is unique. Thus there are at most two lifts of each projected word. The layer and finite-block gates are deterministic and independent of the residue variables. The scalar masked weight of each lift equals the projected weight. Taking the expectation of each signed word first, then its absolute value, therefore puts these words within the majorant of Input 3.2.

Its hypotheses hold with $h = 1$, $C = 101$, $H = \lceil H_0 \rceil$, $Y = B = \lfloor \exp L \rfloor$, $Q_{\max} = \lfloor \exp(100L+1) \rfloor$ and $D_{\max} = \lfloor \exp(2L) \rfloor$. In particular $H_0 \geq \exp(L^{199/200})$, the partial tuple products are at most $\exp(2L)$, and $J + m_Q \leq 101 \log L$. The nonbacktracking state count is at most

$$2(2M_B + 1)\#\mathcal{D} \leq \exp(106L).$$

The at most 2^{2k_*} direction choices, and the factor two for the lift, are absorbed by the $\exp(108L)$ margin in (16), since $\exp(106L)2^{2k_*} \cdot 2 \leq \exp(108L)$ for large L .

For the integer-origin average use Lemma 5.2. Increase ℓ by a multiple of P_* so that $\ell > M_B$. At any initial quotient coordinate $k_0 \in [-M_B, M_B]$, the origins $\ell r + k_0$, $1 \leq r \leq N$, have nonnegative initial origin $a = \ell + k_0$ and step ℓ . Input 3.3 applies with length $N = \exp(L^A)$; there is no bound on ℓ . All actual-word hypotheses hold: $2k_* + 1 \leq 4L$, $2k_*J \leq L^2$, $s \leq L$, $J + m_Q \leq L^2$, and $\#\mathcal{S} \leq \exp(101L)$. Each step has at most $2\#\mathcal{S} \leq \exp(102L)$ numerical choices. The total word and state comparison error is at most

$$\exp(106L + 204L^2 - L^9) = o(1).$$

Combining it with the product-model majorant yields

$$\frac{1}{N} \sum_r \left\| \mathcal{H}_r^{k_*} \right\|_{\text{HS}}^2 \leq 3B_0^{2k_*}.$$

For a finite matrix, $\text{rad}(\mathcal{H}_r)^{2k_*}$ is at most this squared Hilbert–Schmidt norm. Markov therefore shows that $\text{rad}(\mathcal{H}_r) > \exp(3)B_0$ on at most $3N \exp(-6k_*) \leq N \exp(-2k_*)$ origins. At every other origin apply (18) with $a = K$, $b = K(8W)^{J/2}$ and $\text{rad}(\mathcal{H}_r) \leq \exp(3)B_0$. Since $a, b \leq B_0$, it gives $3 \exp(3)B_0 = 3 \exp(3)K(2 \exp(150)\sqrt{W})^J$, which for $J \geq 1$ is at most $K(6 \exp(153)\sqrt{W})^J \leq K(3^{170}\sqrt{W})^J$. This is (36). $\square$

Let C_j° denote (31) after keeping both endpoints according to $\tau_{j,d}$ and E_r .

Proposition 5.4 (The kept and deleted bilinear forms). *For all sufficiently large N ,*

$$|C_j^\circ| \ll \frac{SK(C_3\sqrt{W})^J}{L} + \exp(-L), \quad (38)$$

$$\frac{\sum_j |C_j - C_j^\circ|}{SV_*} \ll 2^J \{K^{-1}(\log L)^{10} + L^{-90} + \exp(-2WJ)\} + \exp(-L^{0.9}). \quad (39)$$

Proof. At origin r use the unweighted tests

$$F_r(k) = \lambda(Nk + r) \mathbf{1}_{\{0 < Nk + r \leq T_j\}}, \quad G_r(t) = \lambda(-Nt - r) \mathbf{1}_{\{0 < -Nt - r \leq N \exp(\eta) D_j\}}.$$

Multiply each by its square-root weight, Q-degree cut and center projection; put f_r in the left layer and g_r in the right layer. All their supports lie in the block because $D_j \leq \exp(100L + \eta)$. Each has $O(D_j)$ supported coordinates. Input 3.3, applied to the truncated weights on the positive and negative physical intervals, gives

$$\sum_r (\|f_r\|^2 + \|g_r\|^2) \ll ND_j S.$$

Cancellation of the square-root weights gives the exact identity

$$\sum_r \left\langle f_r, \left(E_r \sum_d B_{d,r} E_r \right) g_r \right\rangle = LT_j C_j^\circ. \quad (40)$$

Here $t = k - qd$ corresponds to the physical graph site $Nt + r = x - Nqd$. The test reads $Nqd - x = -Nt - r$. All deletion predicates continue to be read at $Nt + r$.

For good origins use (36) and Cauchy–Schwarz in r . At a bad origin bound the pairing row by row on the support of F_r . The weighted absolute row bound is at most $Kw(Nk + r) \sum_d |c_d(Nk + r)|$. On the kept test support, $w(Nk + r) \leq L^{400 \log 5}$ and $\sum_d |c_d(Nk + r)| \leq (8W)^J$. The bad-origin cost is at most $CND_j \exp(-2k_*) KL^{400 \log 5} (8W)^J$. After division by $LT_j = LND_j$ it is $O(\exp(-L))$. This counts only $O(D_j)$ source points per bad origin, and proves (38).

To bound the deletions replace $|c_d(x)|$ by $a_d^+(x)$ and use $|\lambda| \leq 1$. At the source endpoint, the padding/Q-degree, center-degree and prohibited costs are respectively bounded by Lemma 4.3 and Lemma 4.4. For the target endpoint, first translate each individual (d, q) atom by $-Nqd$. The test $q \mid x$ and envelope $a_d^+(x)$ are unchanged by that translation, since qd divides the shift. The resulting atom is the same positive source cost on a translated interval. The finite comparisons permit this arbitrary offset. Sum the two endpoint bounds only after these individual translations. The real normalization T_j differs from the integer length $[T_j] \geq N$ by less than 1. With $\exp(O(L))$ atoms, each bounded by a fixed power of L , this adds at most $\exp(O(L))/N \ll \exp(-L^9)$ in addition to the comparison errors. This proves (39). $\square$

Proposition 5.5 (Centering).

$$\frac{\sum_j |R_j - C_j|}{SV_*} \ll \exp(J) 2^J L^{-s_0}. \quad (41)$$

Proof. Expand the centered product in (31). For every nonempty subset of mean-prime bands, let v be the product of its selected primes and d' the product from the other bands. After putting $x = qd'm$, complete multiplicativity cancels $\lambda(qd')^2$. For each fixed q, d' and bin, the remaining sum, apart from its external factor $u(q)/(qd')$, has the form

$$\frac{1}{Y} \sum_{m \leq Y} \sum_{v \in Z} \frac{c_v}{v} \lambda(m) \lambda(Nv - m), \quad |c_v| \leq 1. \quad (42)$$

The bin restrictions are included in Z and c_v . At most two dyadic pieces suffice to arrange $Z \subset [D, 2D)$, $Y \asymp ND$, and $H_0/2 \leq D \leq \exp(2L)$. Every v is $\exp(L^\gamma)$ -rough. The factor outside (42) is exact, because $Y = T_j/(qd')$.

Write $m = Nk + r$ and define

$$\begin{aligned} F_r(\xi) &= \sum_{k \geq 0, Nk+r \leq Y} \lambda(Nk+r) e(k\xi), \\ G_r(\xi) &= \sum_{1 \leq t \leq \lceil 2D \rceil} \lambda(Nt-r) e(t\xi), \\ B(\xi) &= \sum_{v \in Z} (c_v/v) e(-v\xi). \end{aligned}$$

The integral of $F_r G_r B$ forces $k + t = v$, so (42) is exactly $Y^{-1} \sum_r \int_0^1 F_r G_r B$. One has $|G_r| \ll D$ and $\|F_r\|_2 \|G_r\|_2 \ll D$ by Parseval. Lemma 2.4 with $\nu_0 = \gamma/A$ gives

$$\frac{1}{N} \sum_r |F_r(\xi)| \ll DL^{-1/4000}.$$

The prefix lengths for different r differ by at most one term; a single common prefix and that error justify this application. Its lower-size condition follows from $\log D \geq \frac{1}{2} L^{1-\delta} \geq L^\gamma$ for large L , and $(1-\delta)/3000 > 1/4000$. Also $D \leq \exp(2L) \leq N$. Lemma 4.5, with $\gamma_1 = \gamma$, $\kappa_1 = 1 - \delta$, gives

$$\|B\|_\infty \ll L^{-\gamma}, \quad \int_0^1 |B|^4 \ll D^{-1} L^{-4\gamma}.$$

On $|B| \leq L^{-1-s_0}$, Parseval bounds the convolution by $O(L^{-1-s_0})$. The complementary set has measure at most $CD^{-1} L^{4(1+s_0)-4\gamma}$. There use the r -averaged F_r bound, $|G_r| \ll D$ and the sup norm of B . Since $Y \asymp ND$, its contribution is

$$O(L^{4(1+s_0)-5\gamma-1/4000}) = O(L^{-1-s_0}),$$

because

$$5\gamma + 1/4000 - (5 + 5s_0) = 1/10000 > 0. \quad (43)$$

For each mean-prime subset, the sum of the external positive factors $u(q)/(qd')$ is at most $S \prod_{i \notin I} V_i \leq SV_*$. There are fewer than 2^J such subsets and $O(L \exp J)$ bins. Multiply these counts by L^{-1-s_0} to obtain (41). $\square$

Proof of Theorem 1.1. Sum (38) over the $O(L \exp J)$ bins and divide by SV_* . Since $V_* \geq W^J$, the spectral contribution is at most

$$C \exp(5J) (C_3/\sqrt{W})^J \leq C \exp(-J),$$

since $\exp(6)C_3 < 3^{176} < 10^{88} < \sqrt{W}$. All terms of (39) are $O(\exp(-J))$:

$$2^J K^{-1} (\log L)^{10} = \exp(-(4 - \log 2)J) (\log L)^{10},$$

and $J = \delta \log L/(6W) + O(1)$ with its positive coefficient fixed. The terms $2^J L^{-90}$, $2^J \exp(-2WJ)$ and $\exp(-L^{0.9})$ are smaller. Moreover

$$\exp(J) 2^J L^{-s_0} = O(\exp(-J)), \quad \frac{(2 + \log 2)\delta}{6W} < s_0,$$

so (41) has the same bound. Use (24) and (32) to conclude

$$\frac{|C(N)|}{N} \ll \exp(-J) \leq \exp(1) (\log N)^{-c_*}, \quad c_* := \frac{\delta}{6WA}.$$

The constants A, W are exactly (19). With the numerical choices in (19),

$$c_* = (6 \cdot 10^{191})^{-1} > 10^{-192} > 10^{-200}.$$

This proves (2). Lemma 2.2, with the modulus-one character, bounds the linear sum in (3) by $O(N(\log N)^{-1/40})$. Insert that estimate and the correlation bound in Lemma 1.2; the difference between $(N-1)/4$ and $N/4$ is absorbed by the error. This proves (1). Choose N_0 so large that, for $N \geq N_0$, one has $J \geq 1$; $L = (\log N)^{1/A}$ exceeds the thresholds of Inputs 3.2–3.4 and 4.2, with $h = 1$ or $h = N$ as indicated, $C = 101$, $W = 10^{180}$ and $E = \emptyset$; of Lemma 4.5, with $\gamma_1 = \gamma$ and $\kappa_1 = 1 - \delta$; and of the finitely many inequalities for large L . Also require N to exceed the thresholds of Lemmas 2.2 and 2.4, the latter with $\nu_0 = \gamma/A$, and of Lemma 2.3 with $\nu = \nu_0/2$. The input thresholds precede the quantification of h and the prime families. All thresholds depend only on $A, W, \delta, \gamma, s_0, C_3$; (21) and (26) are uniform in the prime factorisation of N . Thus one absolute N_0 works for both parities and all four signs. The condition $J \geq 1$ alone requires $\log \log N \geq 6 \cdot 10^{191}$. $\square$

Part II

Consequences of the zero-free half-plane

6 Theorem substitutions

Besides the Lean declaration transcribed in Input 2.1, this part uses Pintz’s explicit formula [28], the zero-density estimate of Chen–Gupta–Li [7], a preprint at the time of writing, Montgomery–Vaughan’s Theorem 13.12 [20], and Martin’s Proposition 3 [16]. Their applications here are written mathematics, independent of the graph argument in Part I. The exponents are tracked explicitly; implied constants retain the dependencies indicated by their subscripts. Theorems 6.1 and 6.2 need some fixed zero-free half-plane $\operatorname{Re} s > 1 - \delta_0$, with $\delta_0 > 0$; the exponents 8 in Theorems 6.3 and 6.4, and consequently 24 for primitive roots, use the specific value $7/8$.

6.1 The binary Goldbach exceptional set

Let $E(X)$ count the even integers $4 \leq m \leq X$ which are not a sum of two primes.

Theorem 6.1. *For every $\varepsilon > 0$,*

$$E(X) \ll_{\varepsilon} X^{3/5+\varepsilon}. \quad (44)$$

Here is the exact major-arc input used. In Pintz, *A new explicit formula in the additive theory of primes with applications I*, Theorem 1 and equation (2.10) of version 1 [28], use its notation

$$X_1 = X^{1-\varepsilon_0}, \quad S_X(\alpha) = \sum_{X_1 < p \leq X} \log p \, e(p\alpha), \quad Q = X/P,$$

with the disjoint major arcs $|\alpha - a/q| \leq 1/(qQ)$, $q \leq P$, $(a, q) = 1$. Write

$$R_1(m) = \int_{\mathfrak{M}} S_X(\alpha)^2 e(-m\alpha) d\alpha.$$

Let $\mathfrak{m} = [1/Q, 1 + 1/Q] \setminus \mathfrak{M}$ and put $R_2(m) = \int_{\mathfrak{m}} S_X(\alpha)^2 e(-m\alpha) d\alpha$. Orthogonality gives $R_1(m) + R_2(m) = \sum_{p+p'=m, p, p' \in (X_1, X]} \log p \log p'$. We use (46) for even $m \in [X/2, X]$. For a small fixed

$\eta > 0$, $P_0 \leq X^{4/9-\eta}$, a sufficiently large fixed H and sufficiently large X , it gives a single $P \in [P_0 X^{-\eta}, P_0]$ such that, for all $m \leq X$,

$$R_1(m) = \sum_{(\rho_i, \chi_i), (\rho_j, \chi_j) \in \mathcal{E}} A(\rho_i) A(\rho_j) \mathfrak{S}(\chi_i, \chi_j, m) \frac{\Gamma(\rho_i) \Gamma(\rho_j)}{\Gamma(\rho_i + \rho_j)} m^{\rho_i + \rho_j - 1} + O_\eta(\mathfrak{S}(m) X \exp(-c_\eta H)) + O_\eta(X^{1-\varepsilon_0}). \quad (45)$$

The set $\mathcal{E}$ in its equation (2.8) consists of the pole $(1, \chi_0 \pmod{1})$ with $A(1) = 1$, and nontrivial zeros of primitive characters of conductor at most P , with $\beta \geq 1 - H/\log X$, $|\gamma| \leq \sqrt{X}$, and $A(\rho) = -1$. All small constants, including $\varepsilon_0 > 0$, are fixed according to the major-arc range; the allowed major-arc parameters also satisfy $(\log X)^C \leq P \leq \sqrt{X}$. The minor-arc input is [28, (5.2)], obtained from Parseval's identity and Vaughan's estimate:

$$|R_2(m)| \leq X/\sqrt{\log X} \quad (46)$$

for all $m \leq X$ except $\ll (\log X)^{10} \max(X/P, X^{3/5})$ values. For $P \leq X^{2/5}$ this is the estimate quoted as (2.6) in [27].

Proof. Fix $0 < \eta < \min(\varepsilon/2, 2/45)$ and take $P_0 = X^{2/5}$. Choose $H \geq H_0(\eta)$ first so that the first implied constant in (45) times $\exp(-c_\eta H)$ is less than $1/8$. Then take X sufficiently large for the theorem and $\log X > 8H$. Input 2.1 leaves only the pole in $\mathcal{E}$, and hence

$$R_1(m) = \mathfrak{S}(m)m + O_\eta(\mathfrak{S}(m)X \exp(-c_\eta H)) + O_\eta(X^{1-\varepsilon_0}).$$

For even m , $\mathfrak{S}(m) \geq 2 \prod_{p \geq 3} (1 - (p-1)^{-2}) > 0$. Thus $\operatorname{Re} R_1(m) \geq \mathfrak{S}(m)X/4 \gg X$ for $X/2 \leq m \leq X$. Outside the exceptional set in (46), $R_1(m) + R_2(m) > 0$, which counts two actual primes with positive logarithmic weights. Since $P \geq X^{2/5-\eta}$, the exceptions in this dyadic interval number at most $C_\eta X^{3/5+\eta}(\log X)^{10}$. Absorb the logarithms into $X^{\varepsilon-\eta}$ and sum the dyadic intervals to prove (44). $\square$

The exponent $3/5$ is Pintz's: [28, Theorem 2, (2.13)–(2.14)] obtains it under the stated Siegel-zero hypothesis, and [28, Theorem 3, (2.16)] under a zero-free box adjacent to $\sigma = 1$. Its value comes from the minor-arc estimate [28, §5]. Input 2.1 makes this conditional exponent unconditional by leaving only the pole in (45). The argument uses only a fixed zero-free half-plane. For the preceding unconditional bounds, Pintz [27, Theorem 1] obtained 0.72, and Zhao's preprint [35, Theorem 1.1, version 2] states 0.7. The generalized-zero cutoff in (45) is scaled by $\log X$.

6.2 The least prime in a reduced progression

For $(a, q) = 1$ let $p(a, q)$ be the least prime congruent to $a \pmod{q}$.

Theorem 6.2. *For every $\varepsilon > 0$, uniformly for positive integers q and all reduced residue classes $a \pmod{q}$,*

$$p(a, q) \ll_\varepsilon q^{7/3+\varepsilon}. \quad (47)$$

The proof uses Proposition A.1, which derives the needed case directly from Chen–Gupta–Li's Theorem 1.2. For comparison, Corollary 1.4, general-modulus clause in version 2 of *Large Value Estimates for Dirichlet Polynomials with Characters and Zero Density of Dirichlet L-Functions* [7], assumes $h \leq x$, $q_1 \mid q$, $q_1 \geq \sqrt{q}$, $(a, q) = 1$, and a common zero-free region $\sigma \geq 1 - \eta(q, T)$, $|\Im s| \leq T$ for $\prod_{\chi \pmod{q}} L(s, \chi)$, with

$$\eta(q, x^{\varepsilon/3} T_0) \log x \longrightarrow \infty \quad \text{uniformly for } q < h/x^{17/30-\varepsilon}, \quad T_0 = (x/h)^{1+o(1)}.$$

With the usual meromorphic interpretation at the principal pole, its conclusion is

$$\psi(x+h; q, a) - \psi(x; q, a) = (1 + o_{\eta, \epsilon}(1))h/\varphi(q)$$

provided

$$h/\varphi(q) \geq x^{1/2+\epsilon}q^{1/6} + x^{17/30+\epsilon}.$$

The paper states this corollary with a reference to a smoothed explicit formula. Proposition A.1 supplies that deduction from Theorem 1.2 for the case used here. Input 2.1 supplies its zero-free hypothesis at every modulus and height with $\eta(q, T) = 1/9$; thus $\eta(q, x^{\epsilon/3}T_0) \log x \rightarrow \infty$ uniformly in q .

Proof. Given $\varepsilon > 0$, take $\delta = \min(1/200, \varepsilon/100)$ and fix a nonzero nonnegative $\Phi \in C_c^\infty((1, 2))$. For $q \leq x^{3/7-\delta}$, Proposition A.1 gives a prime $x < p < 2x$ in every reduced class for $x \geq x_0(\delta, \Phi)$. Set $x = q^{1/(3/7-\delta)}$. Since

$$\frac{1}{3/7-\delta} - \frac{7}{3} = \frac{49\delta}{9-21\delta} \leq 6\delta < \varepsilon,$$

we obtain (47); the finitely many smaller moduli, including $q = 1$, are absorbed in its constant. $\square$

The previous published exponent is 5, due to Xylouris [33, Theorem 2.1], with an effective constant; [34, Theorem 1] is a published condensed version and states $p(a, q) < Cq^5$. Zhao [35, Theorem 1.4] also states 5. Naslund's computer-assisted preprint [21] states $p(a, q) < q^{3.99}$ for all sufficiently large q , uniformly in a .

6.3 The least quadratic non-residue

For an odd prime p let $n(p)$ be its least positive quadratic non-residue.

Theorem 6.3. *Uniformly for all odd primes,*

$$n(p) \ll (\log p)^8. \tag{48}$$

The input is Montgomery–Vaughan, *Multiplicative Number Theory I*, Theorem 13.12 [20], pages 428–430. For a nonprincipal character $\chi \pmod{q}$ it assumes $1/\log q \leq \delta \leq 1/2$ and

$$L(s, \chi) \neq 0 \quad \text{for } 1 - \delta < \sigma < 1, \quad 0 < |t| \leq \delta^2 \log q.$$

Its conclusion is $n(\chi) < (A_{\text{MV}}\delta \log q)^{1/\delta}$ for an absolute constant A_{MV} , where $n(\chi)$ is the least n with $\chi(n) \notin \{0, 1\}$.

Proof. Take χ to be the Legendre character modulo p and $\delta = 1/8$. For $p \geq \exp(8)$ the parameter range holds, and Input 2.1 supplies the stated zero-free rectangle. Thus $n(p) < (A_{\text{MV}}/8)^8 (\log p)^8$; the remaining odd primes are covered by $n(p) < p$ and an absolute enlargement of the constant. $\square$

The classical unconditional comparison is Burgess [3, Theorem 2]: $n(p) \ll_{\epsilon} p^{1/(4\sqrt{\exp(1)})+\epsilon}$. OpenAI's accompanying manuscript already gives $n(p) \leq C(\log p)^A$, with $A = 32$ as an example in [25, Corollary 1.2], via Bhargava–Ivanyos–Mittal–Saxena. The companion [24] uses Montgomery–Vaughan's Theorem 13.12 for a fixed power of $\log p$. Equation (48) gives its explicit exponent $1/\delta = 8$ at $\delta = 1/8$.

6.4 Primitive roots and generators

For a modulus q whose unit group is cyclic, let $g(q)$ and $g^*(q)$ be the least positive integer and least prime primitive roots, respectively.

Theorem 6.4. *For every odd prime p , with $r = \omega(p-1)$,*

$$g(p) \leq g^*(p) \ll r^{16}(1 + \log r)^{16}(\log p)^8 \ll (\log p)^{24}. \quad (49)$$

For every $q \geq 3$ with cyclic unit group the same assertions hold with $r = \omega(\varphi(q))$ and p replaced by q , with absolute constants. These moduli are 4, odd prime powers, and twice odd prime powers. For $q = 2$ one has $g(2) = 1$ and $g^(2) = 3$.*

The input is Martin, *The least prime primitive root and the shifted sieve*, Proposition 3 [16]. Write $E(q)$ for the exponent of the unit group, $s(m)$ for the largest squarefree divisor of m , and $\Phi^*(q) = \{\chi^{E(q)/s(\varphi(q))} : \chi \pmod{q}\}$. Let c_0 be the proportion of unit classes having order $E(q)$ and $\log_1 u = \max(\log u, 1)$. The printed conditions of Proposition 3 are $q \geq 2$ and $1/2 \leq \sigma < 1$. We use it for $q \geq 3$, so $\omega(\varphi(q)) \geq 1$, and put

$$f(q, \sigma) = \{\omega(\varphi(q))^2 \log_1 \omega(\varphi(q)) c_0^{-1} \log q\}^{1/(1-\sigma)}.$$

Martin's hypothesis is that every nonprincipal $\chi \in \Phi^*(q)$ has no zero with $\operatorname{Re} s > \sigma$ and $|\Im s| < f(q, \sigma)$. His conclusion is $g^*(q) \ll_\sigma f(q, \sigma)$, where for general q his $g^*(q)$ denotes the least prime of maximal order. His Lemma 6 gives the smoothed estimate

$$\sum_{n \leq x} \Lambda(n) \chi(n) (1 - n/x) \ll x^\sigma \log q$$

for those nonprincipal characters when $1 \leq x \ll T \ll q$ and their L -functions have no zero with $\operatorname{Re} s > \sigma$, $|\Im s| < T$. At the polylogarithmic search length below one may take $T = x \ll q$.

Proof. Input 2.1 supplies Martin's hypothesis with $\sigma = 7/8$ at every height. For a cyclic group, $c_0 = \varphi(\varphi(q))/\varphi(q)$. Put $n = \varphi(q)$ and let p_i be the i th prime. Mertens' theorem and $p_r \ll r \log(2r)$ give

$$c_0^{-1} = \frac{n}{\varphi(n)} \leq \prod_{1 \leq i \leq \omega(n)} (1 - 1/p_i)^{-1} \ll \log_1 \omega(n),$$

as in Martin's remark after Proposition 3. Thus $f(q, 7/8) \ll r^{16}(1 + \log r)^{16}(\log q)^8$. If $r \geq 2$, the product of r distinct prime divisors of $\varphi(q)$ is at least $r!$, so $r(1 + \log r) \ll \log q$ by the factorial bound; the case $r = 1$ follows since $q \geq 3$. Substitution yields $(\log q)^{24}$, and $g(q) \leq g^*(q)$ is immediate. $\square$

Corollary 6.5 specializes Bach's character and proper-subgroup bound [1, p. 293, unnumbered theorem and the subsequent subgroup reduction] to the uniform zero-free boundary 7/8.

Corollary 6.5. *There is an absolute $C > 0$ such that for every $q \geq 2$ and every nonprincipal character $\chi \pmod{q}$ there is a prime $\ell \nmid q$, $\ell < C(\log q)^8$, with $\chi(\ell) \neq 1$. The unit group modulo q is generated by the primes in this range. For every $k \geq 2$ whose k th-power subgroup is proper, the least positive unit outside it is prime and less than $C(\log q)^8$, with the same constant independent of k .*

Proof. Apply Montgomery–Vaughan’s Theorem 13.12, as stated above, with $\delta = 1/8$ to any non-principal χ . It gives $n(\chi) < C(\log q)^8$ for $q \geq \exp(8)$; finitely many smaller moduli are included by enlarging C . The least such unit $n(\chi)$ is prime: if it were composite, all its prime factors would be smaller units on which χ equals 1. If the indicated primes generated a proper subgroup, the quotient would have a nontrivial character, contradicting this conclusion. A proper k th-power subgroup also has a nontrivial annihilating character. Hence a prime in the stated range lies outside it; the least positive unit outside a subgroup is prime by the same factorisation argument. $\square$

Sandlund [30], in a research memo dated 7 October 2026, derives from a Dirichlet zero-free half-plane $\operatorname{Re} s > 1 - \delta$ the bound $K_\delta(\log 2m)^{1/\delta}$ for a unit prime outside a proper subgroup of the units modulo m and, via Shoup’s sieve, the bound $C_\delta\{k^2(1 + \log k)^2 \log 2p\}^{1/\delta}$, $k = \omega(p-1)$, for the least prime primitive root, and constructs primitive roots when the factorisation of $p-1$ is supplied. With $\delta = 1/8$ these are the bounds of Corollary 6.5 and, for prime moduli, of Theorem 6.4; that memo precedes the present paper. The proof of Theorem 6.4 given here goes through Martin’s shifted sieve. Two notes of 8 October 2026 in the zeta-lab repository [32] state the exponent $7/3 + \varepsilon$ of Theorem 6.2 with an effective constant, and the bound $n(\chi) \leq (\log q)^8$ for every nonprincipal character modulo $q \geq 3$.

The classical unconditional bound $g(p) \ll_\varepsilon p^{1/4+\varepsilon}$ is due to Wang in 1959, using Burgess’s character-sum estimate, as credited in [2, §1] and [10, §1]; the same exponent follows from Burgess [4, Theorem 1]. Indeed that theorem gives a power saving for nonprincipal character sums of length $H = \lceil p^{1/4+\varepsilon} \rceil$ on choosing its integer parameter larger than $1/(4\varepsilon)$; the primitive-root character expansion then uses $2^{\omega(p-1)} = p^{o(1)}$ to give the stated bound. Carella [6, Theorem 1.1] claims the unconditional bound $g(p) \ll_\varepsilon (\log p)^{1+\varepsilon}$ for fixed small $\varepsilon > 0$ and sufficiently large primes. For the least prime primitive root, Ha’s publisher abstract [11] reports the unconditional exponent 3.1. Martin [16, Theorem 1] obtains a fixed power of $\log p$ for all but $O_{\varepsilon,\eta}(Y^\varepsilon)$ odd prime powers $p^t \leq Y$, with $0 < \varepsilon \leq 20/21$ and $\eta > 0$. Under GRH, Wang’s bound, as recorded in [31, §1], is $g(p) \ll r^6(\log p)^2$. Shoup [31, Theorem 3] obtains $g(p) \ll r^4(1 + \log r)^4(\log p)^2 \ll (\log p)^6$; Martin [16, Corollary 3.1] states the prime version and extends it to composite moduli as an estimate for primes of maximal order. Bach [2, Theorem 3.2], under GRH and for odd primes p , constructs a set of $O((\log p)^4/(\log \log p)^3)$ residue classes containing a primitive root, using $O((\log p)^7/(\log \log p)^3)$ bit operations with ordinary arithmetic. The bound is for the number of candidate classes.

A A uniform density deduction for progressions

Proposition A.1. *Fix $0 < \delta < 1/100$ and a nonzero nonnegative $\Phi \in C_c^\infty((1, 2))$. Uniformly for $q \leq x^{3/7-\delta}$ and $(a, q) = 1$, as $x \rightarrow \infty$,*

$$\sum_{n \equiv a \pmod{q}} \Lambda(n) \Phi(n/x) = (c_\Phi + o_{\delta, \Phi}(1)) \frac{x}{\varphi(q)}, \quad c_\Phi = \int_1^2 \Phi(u) du > 0.$$

Every such class contains a prime at most $2x$ once x is sufficiently large in terms of δ, Φ .

The input is Chen–Gupta–Li, *Large Value Estimates for Dirichlet Polynomials with Characters and Zero Density of Dirichlet L-Functions*, Theorem 1.2, last universal estimate in version 2 [7]. For $q \geq 1$ and $1/2 < \sigma < 1$, with $N(\sigma, T, \chi)$ counting zeros $\beta + i\gamma$ with $\sigma \leq \beta \leq 1$, $|\gamma| \leq T$, the final “in all cases” display of Theorem 1.2 states, with the paper’s convention $qT \rightarrow \infty$ for $o(1)$ (we use $T = x^\tau \rightarrow \infty$),

$$\sum_{\chi \bmod q} N(\sigma, T, \chi) \leq (qT)^{o(1)} \{(q^{7/3} T^2)^{1-\sigma} + (qT)^{(30/13)(1-\sigma)}\}. \quad (50)$$

This clause has no smoothness condition on q . Its $o(1)$ means that any fixed positive power of qT is an admissible loss with a corresponding fixed implied constant.

Proof. Fix $0 < \delta < 1/100$ and suppose $q \leq x^{3/7-\delta}$. Take $\tau = \delta/8$, $T = x^\tau$, and density loss $\alpha = \delta/100$ in (50). Then

$$q^{7/3}T^2 \leq x^{1-25\delta/12} \leq x^{1-\delta}, \quad (qT)^{30/13} \leq x^{90/91-105\delta/52} \leq x^{1-\delta}.$$

Fix a nonzero nonnegative $\Phi \in C_c^\infty((1, 2))$, with Mellin transform $\tilde{\Phi}$ and $c_\Phi = \tilde{\Phi}(1) > 0$. The smoothed explicit formula, from [20, Theorem 12.10 and the zeta formula in Chapter 12], and character orthogonality give

$$\sum_{n \equiv a \pmod{q}} \Lambda(n) \Phi(n/x) = \frac{c_\Phi x}{\varphi(q)} + O_\Phi \left(\frac{x}{\varphi(q)} \sum_{\chi \bmod q} \sum_{\rho} x^{\beta-1} |\tilde{\Phi}(\rho)| + \log^2(qx) \right).$$

For primitive χ , [20, Theorem 12.10] and its zeta analogue, integrated by parts against $\Phi'(u/x)/x$, give

$$\sum_n \Lambda(n) \chi(n) \Phi(n/x) = \mathbf{1}_{\chi=\chi_0} \tilde{\Phi}(1)x - \sum_{\rho} x^\rho \tilde{\Phi}(\rho) + O_\Phi(1).$$

The constant $C(\chi)$ in that theorem's (12.6) is independent of u and integrates to zero against Φ' . For fixed x, q , its remainder tends to zero after integration as the truncation height tends to infinity: the printed remainder bound gives pointwise convergence away from integers and an integrable bound on this compact interval. The zero series converges absolutely by $\tilde{\Phi}(\beta + i\gamma) \ll_{\Phi, B} (1 + |\gamma|)^{-B}$. The induced Euler factors cost $O_\Phi(\log^2(qx))$ after orthogonality; the principal pole supplies the main term.

All nontrivial zeros have $\beta \leq 7/8$. For $|\gamma| \leq T$ and $\beta \leq 1/2 + \alpha$, the unit-height count [20, Theorem 10.17] bounds the normalized zero cost by $O_\Phi(qT \log(qT)x^{-1/2+\alpha}) = o(1)$. Partition $[1/2 + \alpha, 7/8]$ into intervals of width at most α . For each lower endpoint σ , (50) bounds its cost by

$$C_{\delta, \Phi} x^{\sigma+\alpha-1} x^\alpha x^{(1-\delta)(1-\sigma)} \leq C_{\delta, \Phi} x^{-\delta/8+2\alpha} = o(1).$$

This uses only finitely many fixed values of σ . For $|\gamma| > T$, choose $B = \lceil 10/\tau \rceil$ in the rapid decay $\tilde{\Phi}(\beta + i\gamma) \ll_{\Phi, B} (1 + |\gamma|)^{-B}$. The aggregate tail is $O_{\delta, \Phi}(qx^{-1/8}T^{1-B} \log(qT)) = o(1)$. Thus the weighted progression sum is $(c_\Phi + o(1))x/\varphi(q)$, uniformly in the stated range and in a . Prime powers of exponent at least two cost $O(\sqrt{x} \log^2 x) = o(x/\varphi(q))$, since $x/\varphi(q) \geq x/q \geq x^{4/7+\delta}$. There is therefore a prime at most $2x$ in every such class. This proves the uniform asymptotic and the prime assertion. $\square$

The same progression range follows from any fixed zero-free half-plane $\operatorname{Re} s > 1 - \delta_0$, with $0 < \delta_0 \leq 1/2$. Take $\alpha = \min(\delta/100, \delta\delta_0/10)$, partition up to $1 - \delta_0$, and replace the grid cost by $x^{-\delta\delta_0+2\alpha} = o(1)$ and the tail factor by $x^{-\delta_0}$. The other bounds above are unchanged.

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Use of AI

The proofs and the text of this paper were produced with AI systems (GPT-6 Astra, GPT-6.1 Sol and Claude Opus 5.5) under the author’s direction. They were checked by independent reviews with AI models; the inputs from the OpenAI library are Lean theorems, and Theorem 1.1 is formalized in Lean 4 with one published estimate as hypothesis. The author takes responsibility for the content.